

WireGuard VPN Setup Guide for AWS Windows Server and Client PC

About the Author

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Introduction

This document provides the configuration details for setting up a WireGuard VPN between an AWS Windows Server and client PCs. The setup includes restricting RDP access to only VPN-connected clients.

Server Configuration (AWS Windows Server)

Example Server Configuration File (NTC_SERVER.conf)

[Interface]

PrivateKey = <server_private_key>

ListenPort = 51820

Address = 10.0.0.1/24

[Peer]

PublicKey = <client_public_key>

AllowedIPs = 10.0.0.2/32

Key configuration parameters:

- ListenPort: 51820 (ensure this port is open in AWS security groups)

- Address: Server's VPN IP address (10.0.0.1 in this example)
- Peer PublicKey: Client's public key
- AllowedIPs: Client's VPN IP address (10.0.0.2 in this example)

Client Configuration

Example Client Configuration File (NTC_CLIENT.conf)

[Interface]

PrivateKey = <client_private_key>

Address = 10.0.0.2/24

DNS = 8.8.8.8, 1.1.1.1

[Peer]

PublicKey = <server_public_key>

AllowedIPs = 0.0.0.0/0

Endpoint = <server_public_ip>:51820

PersistentKeepalive = 25

Key configuration parameters:

- Address: Client's VPN IP address (10.0.0.2 in this example)
- DNS: DNS servers to use while connected (Google and Cloudflare DNS in this example)
- Peer PublicKey: Server's public key
- Endpoint: Server's public IP and port (e.g., 203.0.113.45:51820)
- PersistentKeepalive: Helps maintain connection through NAT (25 seconds in this example)

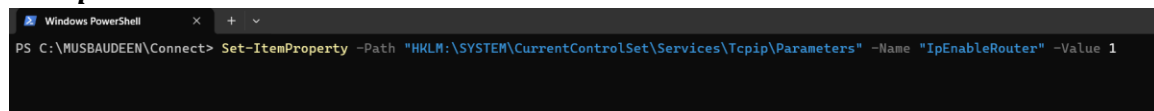
Enable IP Forwarding (Optional)

- If you want the server to route traffic for clients:

Open PowerShell as Administrator and run: ***Set-ItemProperty -Path***

"HKLM:\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters" -Name

"IpEnableRouter" -Value 1



```
Windows PowerShell
PS C:\MUSBAUDEEN\Connect> Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters" -Name "IpEnableRouter" -Value 1
```

Restart-Computer

Security Configuration

After VPN setup, RDP access should be restricted to only the VPN subnet:

AWS Security Group Rules

- Inbound RDP (3389) access should be limited to the VPN subnet (10.0.0.0/24)
- WireGuard (UDP 51820) access should be allowed from client IPs or specific networks

Example Key Generation

Example of how keys are generated (run these commands on respective machines):

Generate private key

```
wg genkey > private.key
```

Generate public key from private key

```
wg pubkey < private.key > public.key
```

Troubleshooting

- Verify keys match between server and client configurations
- Check AWS security group rules for proper ports and IP restrictions
- Confirm the server's public IP hasn't changed (affects client Endpoint)
- Test basic connectivity (ping between VPN IPs) before testing RDP